# INORGANIC CUMULATIVE EXAM

**April 25, 1998**

## Inorganic Reactions

Write a balanced equation for a reaction which will go in the direction written (use specific formulas for reactants and products, describe geometries where necessary, give necessary catalysts and conditions, etc.) to illustrate each of the following types of inorganic reactions:

1. A photochemical reaction of an organometallic complex.
2. A reductive-elimination reaction of an organometallic complex.
3. An oxidative-addition reaction of an organometallic complex (not the reverse of #2)
4. A carbonyl insertion reaction.
5. A reaction controlled by the trans effect.
6. An inner sphere electron transfer reaction.
7. A inorganic polymer-producing reaction.
8. A linkage isomerism reaction.
. A Lewis acid/Lewis base reaction involving antimony.
10. A metathesis reaction controlled by hard-soft considerations.
11. A racemization reaction of an octahedral complex.
12. A reaction of xenon.
13. A disproportionation reaction.